

Question:

Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Understand the problem

Josh buys a house for \$ 80 , 000 and spends \$ 50 , 000 on repairs . The repairs increase the value of the house by 150 % . We are to find the profit he made .

Step 2 : Find the new value of the house after repairs

The value increases by 150 % . So we calculate :

Increase = 150 % of \$ 80 , 000 = (150 / 100) Ã 80 , 000 = 1.5 Ã 80 , 000 = \$ 120 , 000

So the new value of the house is :

New value = Original value + increase = 80 , 000 + 120 , 000 = \$ 200 , 000

Wait â this interpretation is incorrect . The problem says "the repairs increased the value of the house by 150 % " . But â "150 % increase" means the new value is 150 % more than the original value . So the original value is \$ 80 , 000 , and the value is now increased by 150 % of that .

So new value = 80 , 000 + (150 % of 80 , 000) = 80 , 000 + (1.5 Ã 80 , 000) = 80 , 000 + 120 , 000 = \$ 200 , 000

But wait â this is only if the value increases by 150 % from the original . However , the cost of repairs is \$ 50 , 000 â which is a cost , not the original value . Is the 150 % increase applied to the original value or the total cost ?

The problem says : " He buys a house for \$ 80 , 000 and puts in \$ 50 , 000 in repairs . This increased the value of the house by 150 % . "

The 150 % increase is on the original value â as it says "the value increased by 150 % " , not "the cost increased " , etc .

Therefore , the value of the house after repairs becomes 150 % more than the original value .

So new value = 80 , 000 + (150 % of 80 , 000) = 80 , 000 + 120 , 000 = \$ 200 , 000

But that would mean the value went from \$ 80 , 000 to \$ 200 , 000 . Then the profit would be :

Total cost to Josh = cost of buying + cost of repairs = 80 , 000 + 50 , 000 = \$ 130 , 000

Profit = Selling value - Total cost = 200 , 000 - 130 , 000 = \$ 70 , 000

Therefore , the profit is \$ 70 , 000 .

But wait â let's verify : is the 150 % increase based on the original value or on the total cost ? It says "increased the value of the house by 150 % " , which means 150 % of original value .

So yes , original value = \$ 80 , 000 â new value = 80 , 000 Ã (1 + 1.5) = 80 , 000 Ã 2.5 = \$ 200 , 000

Total cost = 80 , 000 + 50 , 000 = 130 , 000

Profit = 200 , 000 - 130 ,